

Linear Algebra - CS - 3A & 3B

Problem Sheet

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Linear Combinations

1. In computer vision, three feature vectors extracted from an image are:

$$\mathbf{f}_1 = (2, 1, 3), \quad \mathbf{f}_2 = (-1, 2, 1), \quad \mathbf{f}_3 = (4, -1, 2)$$

A new image produces feature vector $\mathbf{g} = (7, 3, 8)$. Can $\mathbf{g}$ be expressed as a linear combination of $\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3$? If so, find the coefficients. What does this indicate about the features' ability to represent new images?

2. Three color palettes in a graphic design tool are represented as RGB vectors:

$$\mathbf{p}_1 = (255, 0, 0), \quad \mathbf{p}_2 = (0, 255, 0), \quad \mathbf{p}_3 = (0, 0, 255)$$

A designer wants to create a new color $\mathbf{c} = (128, 128, 255)$ by mixing these palettes. Find the mixing ratios. Is this combination unique?

3. In audio processing, three sound waves are represented by amplitude vectors:

$$\mathbf{w}_1 = (1, 2, -1), \quad \mathbf{w}_2 = (2, -1, 3), \quad \mathbf{w}_3 = (-1, 3, 2)$$

A composite wave $\mathbf{s} = (4, 5, 5)$ is detected. Decompose $\mathbf{s}$ into a linear combination of $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$. Interpret the coefficients as volumes of each source wave.

4. Three basis vectors in a 3D rendering engine are:

$$\mathbf{b}_1 = (1, 1, 0), \quad \mathbf{b}_2 = (1, 0, 1), \quad \mathbf{b}_3 = (0, 1, 1)$$

A vertex is located at $\mathbf{v} = (3, 2, 2)$ in standard coordinates. Express $\mathbf{v}$ in terms of the basis $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$. If the basis is orthogonalized, would the coordinates change?

Spanning/Generating vectors

5. A robot arm moves in 3D space with three joints. The reachable positions are all linear combinations of:

$$\mathbf{a}_1 = (2, 1, 0), \quad \mathbf{a}_2 = (1, 2, 1), \quad \mathbf{a}_3 = (3, 3, 1)$$

Does the set $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ span $\mathbb{R}^3$? If not, describe the subspace of reachable positions. Can the robot reach position $(1, 1, 2)$?

6. In data science, three feature vectors from a dataset are:

$$\mathbf{d}_1 = (1, 2, 3), \quad \mathbf{d}_2 = (2, 3, 4), \quad \mathbf{d}_3 = (3, 4, 5)$$

Do these vectors span $\mathbb{R}^3$? If not, what is the dimension of their span? A new data point $(4, 5, 6)$ is collected. Is it in the span? What does this imply about adding redundant features?

7. Three force vectors act on a physics simulation:

$$\mathbf{f}_1 = (1, 0, 1), \quad \mathbf{f}_2 = (0, 1, 1), \quad \mathbf{f}_3 = (1, 1, 0)$$

Do these forces span all possible 3D force directions? Find the set of all force directions that cannot be achieved by combining these three forces.

8. A neural network's hidden layer produces outputs that are linear combinations of:

$$\mathbf{h}_1 = (1, -1, 2), \quad \mathbf{h}_2 = (2, 1, -1), \quad \mathbf{h}_3 = (3, 0, 1)$$

Determine whether $\{\mathbf{h}_1, \mathbf{h}_2, \mathbf{h}_3\}$ spans $\mathbb{R}^3$. If not, find a basis for their span. A new sample produces $(4, -1, 4)$. Can the network represent this output?

Linear Independence of Vectors

9. In cryptography, three encryption keys are represented as vectors:

$$\mathbf{k}_1 = (1, 2, 3), \quad \mathbf{k}_2 = (2, 3, 1), \quad \mathbf{k}_3 = (3, 1, 2)$$

Are these keys linearly independent? If they are dependent, find a non-trivial combination that yields zero. How does this affect the security of the encryption scheme?

10. Three network traffic patterns are recorded as vectors of packet counts through three routers:

$$\mathbf{t}_1 = (150, 200, 100), \quad \mathbf{t}_2 = (300, 400, 200), \quad \mathbf{t}_3 = (75, 100, 50)$$

Determine whether these traffic patterns are linearly independent. If they are dependent, explain which pattern can be expressed as a combination of the others and what this means for network monitoring.

11. In machine learning, three feature vectors from a dataset are:

$$\mathbf{x}_1 = (1, 0, 1), \quad \mathbf{x}_2 = (0, 1, 1), \quad \mathbf{x}_3 = (1, 1, 0)$$

Check for linear independence. If they are independent, can they form a basis for $\mathbb{R}^3$? If not, find the dimension of their span and a basis for it.

12. Three color vectors in an RGB color space are

$$\mathbf{c}_1 = (255, 128, 0), \quad \mathbf{c}_2 = (128, 255, 0), \quad \mathbf{c}_3 = (0, 128, 255)$$

Test these vectors for linear independence. If they are dependent, find the relationship. Would removing one vector reduce the color gamut? Explain.